

LAUWERSOOG, Netherlands

WMO: 062770

Lat: 53.4117N

Lon: 6.1992E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.4	(c) -3.9	(d) -8.7	(e) 1.8	(f) -4.3	(g) -6.7	(h) 2.1	(i) -3.4	(j) 18.1	(k) 7.0	(l) 16.6	(m) 6.5	(n) 5.3	(o) 100	(p) 0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.6	(c) 26.3	(d) 19.8	(e) 24.0	(f) 19.0	(g) 22.4	(h) 18.2	(i) 20.7	(j) 24.8	(k) 19.7	(l) 23.0	(m) 18.8	(n) 21.7	(o) 5.0	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.0	22.7	18.3	13.2	21.6	17.6	12.6	20.7	59.9	25.3	56.3	23.1	53.4	21.8	
		</													

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.1	(b) 13.4	(c) 12.1	(e) -6.6	(f) 31.4	(g) 2.5	(h) 1.8	(i) -8.5	(j) 32.7	(k) -9.9	(l) 33.7	(m) -11.4	(n) 34.7	(o) -13.2	(p) 36.0		
(a) 15.1	(b) 13.4	(c) 12.1	(e) -7.3	(f) 23.3	(g) 2.5	(h) 0.9	(i) -9.1	(j) 24.0	(k) -10.6	(l) 24.5	(m) -12.0	(n) 25.0	(o) -13.8	(p) 25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	3.1	3.7	5.9	9.6	13.0	15.8	17.9	18.0	15.6	11.4	7.3	4.2
	DBStd	6.13	3.56	3.32	2.97	3.06	2.89	2.54	2.70	2.33	2.30	2.82	2.88	3.62
	HDD10.0	857	216	178	129	43	4	0	0	0	0	19	88	181
	HDD18.3	2934	474	411	385	263	168	85	41	33	88	216	332	438
	CDD10.0	1033	0	0	2	31	98	173	246	249	167	61	6	1
	CDD18.3	68	0	0	0	1	2	8	28	24	5	0	0	0
	CDH23.3	261	0	0	0	3	11	43	103	83	17	0	0	0
	CDH26.7	56	0	0	0	0	1	11	26	17	2	0	0	0
(14) Wind	WSAvg	6.2	6.9	6.7	6.7	6.2	6.4	6.1	5.9	5.4	5.7	6.2	6.1	6.6
(15) Precipitation	PrecAvg	858	70	57	48	42	56	71	84	93	87	83	85	82
	PrecMax	1049	138	140	104	114	106	122	186	204	206	171	156	190
	PrecMin	595	4	21	7	1	12	28	34	19	12	31	11	27
	PrecStd	125	34	26	27	23	26	31	38	47	41	32	34	39
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.8	15.6	22.4	24.7	28.2	29.7	29.1	25.8	20.4	14.8	12.2
		MCWB	9.8	9.2	10.8	14.3	16.6	20.7	21.0	20.0	19.4	16.7	13.2	11.0
	2%	DB	9.6	10.0	12.4	18.3	21.4	23.7	26.1	25.6	22.5	17.6	13.3	10.9
		MCWB	8.6	8.4	9.5	13.1	16.0	18.8	20.4	19.6	18.1	15.2	12.1	9.9
	5%	DB	8.5	8.8	10.7	16.0	19.2	21.2	23.7	23.3	20.4	16.0	12.0	9.8
		MCWB	7.6	7.5	8.9	12.1	15.4	17.5	19.4	18.9	17.3	14.2	10.7	8.9
	10%	DB	7.6	7.8	9.7	14.0	17.4	19.7	22.1	21.7	19.0	15.0	11.0	8.6
		MCWB	6.7	6.7	8.1	11.2	14.2	16.7	18.5	18.1	16.5	13.5	9.8	7.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	10.0	12.0	15.5	18.5	21.3	22.5	22.0	20.2	17.1	13.6	11.3
		MCDB	10.8	11.0	14.4	20.6	21.8	26.4	27.6	26.0	24.7	19.9	14.6	11.9
	2%	WB	8.7	8.7	10.2	13.7	16.9	19.3	20.9	20.5	18.7	15.6	12.2	10.1
		MCDB	9.4	9.7	11.7	17.0	20.3	22.7	25.1	24.2	21.5	17.2	13.2	10.7
	5%	WB	7.7	7.7	9.2	12.6	15.6	18.1	19.8	19.4	17.8	14.6	11.0	9.0
		MCDB	8.4	8.6	10.6	15.5	18.6	21.0	23.2	22.4	19.9	15.9	11.9	9.7
	10%	WB	6.8	6.8	8.2	11.5	14.5	16.9	18.9	18.5	16.8	13.7	10.0	7.8
		MCDB	7.5	7.7	9.4	13.8	17.2	19.2	21.8	21.2	18.7	14.8	10.8	8.5
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	3.8	4.7	5.6	5.3	5.2	5.3	5.6	5.2	4.7	3.7	3.4
		MCDBR	3.6	5.0	6.8	9.5	8.7	8.6	8.6	8.8	7.8	5.8	4.2	3.5
		MCWBR	3.4	4.0	4.6	5.6	5.1	5.1	4.7	4.7	4.6	4.0	3.5	3.3
	5% WB	MCDBR	3.6	4.6	6.2	8.4	7.5	7.9	7.7	7.4	6.7	5.7	4.1	3.5
		MCWBR	3.6	3.9	4.5	5.4	4.8	5.1	4.5	4.5	4.4	4.3	3.7	3.4
(40) Clear-Sky Solar Irradiance	taub		0.330	0.348	0.376	0.405	0.403	0.401	0.412	0.406	0.388	0.363	0.342	0.322
	taud		2.312	2.336	2.281	2.251	2.307	2.331	2.312	2.344	2.383	2.406	2.346	2.306
	Ebn at Noon		628	744	799	823	846	850	832	814	781	719	608	559
	Edh at Noon		61	81	104	121	121	119	119	110	94	75	59	51
(44) All-Sky Solar Radiation	RadAvg		0.59	1.23	2.50	4.16	5.17	5.41	5.15	4.28	3.01	1.67	0.71	0.45
	RadStd		0.06	0.20	0.25	0.40	0.39	0.38	0.57	0.33	0.29	0.16	0.07	0.05

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page